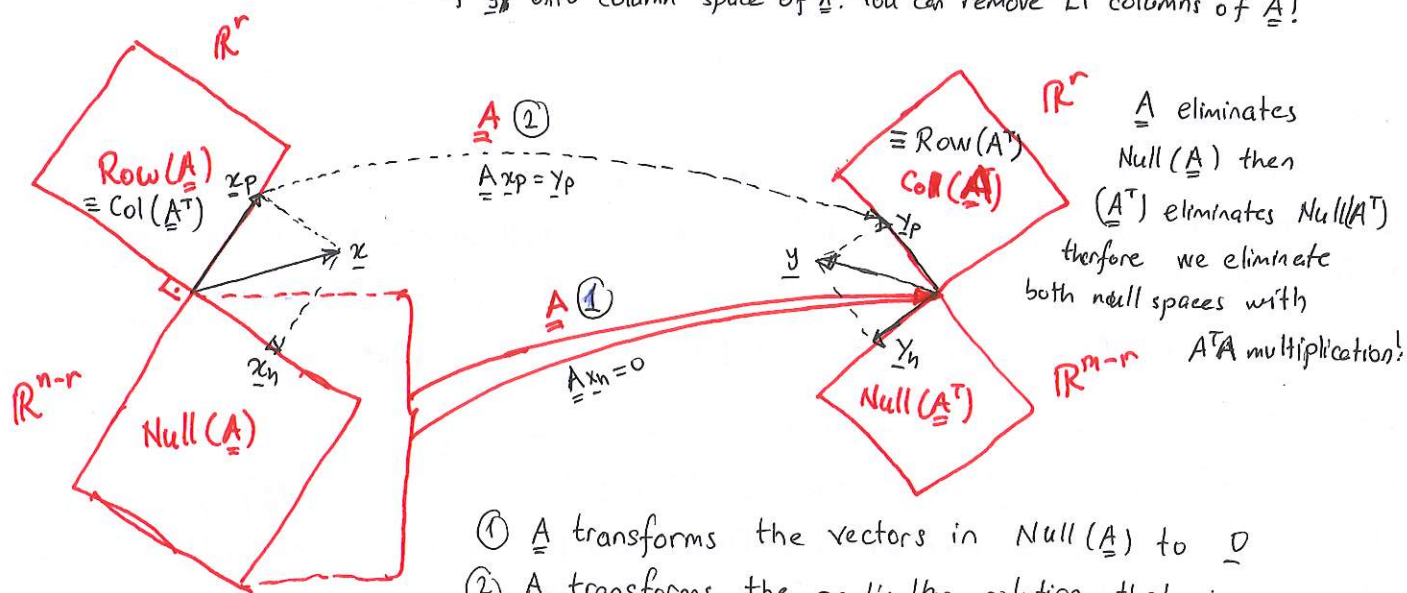


* Project $\underline{y}$ onto column-space of $\underline{A}$: Find $\underline{x}$ for $\underline{A}^T \underline{A} \underline{x} = \underline{A}^T \underline{y}$ then $\underline{y}_p = \underline{A} \underline{x}$, $\underline{y}_p$ is the projection of $\underline{y}$ onto column space of $\underline{A}$. You can remove LI columns of $\underline{A}$!



- ① $\underline{A}$ transforms the vectors in $\text{Null}(\underline{A})$ to $\underline{0}$
- ② $\underline{A}$ transforms the particular solution that is orthogonal to $\text{Null}(\underline{A})$, i.e., that is in $\text{Row}(\underline{A})$ to $\underline{y}_p$.

* There is no

* There is no mapping from domain to $\text{null}(\underline{A}^T)$. For any $\underline{y}$ that has component from $\text{null}(\underline{A}^T)$, i.e., $\underline{y}_n$, there is no solution for the system $\underline{A} \underline{x} = \underline{y}$, since $\underline{y}$ should be in the $\text{col}(\underline{A})$ which is not the case. $\underline{A} \underline{x} = \sum x_i \underline{c}_i \Rightarrow \underline{A} \underline{x} \in \text{Col}(\underline{A})$
 $\sum x_i \underline{c}_i \in \text{Col}(\underline{A})$ for any $\underline{x}$!

* Let $\underline{y} = \underline{y}_p + \underline{y}_n$ then we do not have any solution for the system $\underline{A} \underline{x} = \underline{y}$ since $\underline{A}$ can not map $\underline{x}$ to $\underline{y}_n$ which is in $\text{null}(\underline{A}^T)$.

(*) We can eliminate $\underline{y}_n$ by projecting $\underline{y}$ to $\text{Col}(\underline{A})$. Consider multiplying $\underline{A} \underline{x} = \underline{y}$ with $\underline{A}^T$ from left.

$$\underline{A}^T \underline{A} \underline{x} = \underline{A}^T \underline{y} = \underline{A}^T (\underline{y}_p + \underline{y}_n) = \underline{A}^T \underline{y}_p + \underbrace{\underline{A}^T \underline{y}_n}_{\underline{0} \text{ since } \underline{y}_n \in \text{Null}(\underline{A}^T)} = \underline{A}^T \underline{y}_p \Rightarrow \underline{A} \underline{x} = \underline{y}_p \text{ which is consistent!}$$

- How to project $\underline{y}$ onto column space of $\underline{A}$.

$\underline{y}_p = \underline{A} \underline{x}$ for $\underline{A}^T \underline{A} \underline{x} = \underline{A}^T \underline{y}$: Solve for $\underline{A}^T \underline{A} \underline{x} = \underline{A}^T \underline{y}$ the $\underline{y}_p = \underline{A} \underline{x}$ is the projection of $\underline{y}$ onto the $\text{col}(\underline{A})$.

Example: $\underline{A} = \begin{bmatrix} 1 & 2 & 4 & 3 \\ 1 & 3 & 4 & 4 \\ 1 & 3 & 4 & 4 \end{bmatrix}$, $\underline{B} = \underline{A}^T = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 3 \\ 4 & 4 & 4 \\ 3 & 4 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \underline{M} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$

$$\underline{z} \in \text{Col}(\underline{A}) \Rightarrow \underline{z} = \begin{bmatrix} \underline{I} \\ \underline{M}^T \end{bmatrix} \underline{x} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \underline{x}, \underline{w} \in \text{Null}(\underline{A}^T) \Rightarrow \underline{w} = \begin{bmatrix} 0 & 0 \\ -1 & -1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \underline{\theta} = \begin{bmatrix} \underline{M} \\ \underline{I} \end{bmatrix} \underline{\theta}$$

Let's choose $\underline{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\underline{\theta} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \underline{y}_p = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \in \text{Col}(\underline{A})$

Let us project $\underline{y}$ to $\text{Col}(\underline{A})$ then we should obtain $\underline{y}_p$ $\Rightarrow \text{col}(\underline{B}) = \text{col}(\underline{A})$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \underline{x} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \underline{x} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \underline{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\underline{y}_p = \underline{A} \underline{x} = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \checkmark$$

$$\underline{y}_n = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -1 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \in \text{Null}(\underline{A}^T)$$

$$\underline{y} = \underline{y}_p + \underline{y}_n = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

Let $\underline{x} \in \mathbb{R}^n$ and $\underline{y} \in \mathbb{R}^m$, the matrix $\underline{A}$ maps $\underline{x}$ to $\underline{y}$, i.e., $\underline{y} = \underline{A}\underline{x}$.

* The space that $\underline{x}$ lives in it is called "Domain" and $\underline{y}$ lives in it, is called as the "Range" space.

* If $\underline{x}_h$ is a homogeneous solution the $\underline{A}\underline{x}_h = \underline{0}$. $\underline{x}_h = \underline{0}$ is the trivial solution.

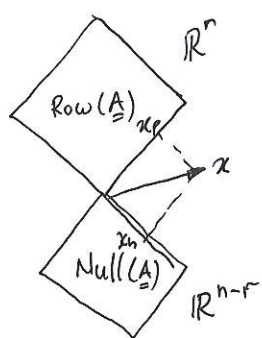
* All possible $\underline{x}_h$ vectors are said to be in null space of $\underline{A}$. ($\text{Null}(\underline{A})$)

* The $\text{null}(\underline{A})$ is orthogonal to row space of $\underline{A}$ ($\text{Row}(\underline{A})$), i.e., $\text{NULL}(\underline{A}) \perp \text{Row}(\underline{A})$

Proof: Let $\underline{r} \in \text{Row}(\underline{A})$ and $\underline{x}_h \in \text{Null}(\underline{A})$, then $\underline{r} = \sum_{i=1}^m \alpha_i \underline{r}_i = \underline{A}^T \underline{\alpha}$, $\underline{r}_i$ is the i th

row of $\underline{A}$ and $\underline{A}\underline{x}_h = \underline{0} \Rightarrow \underline{r} \perp \underline{x}_h$ then $\underline{r}^T \underline{x}_h = 0 \Rightarrow \underline{\alpha}^T \underline{A} \underline{x}_h = \underline{\alpha}^T \underline{0} = 0$

* If the # of rows of $\underline{A}$ is r the $\dim(\text{Row}(\underline{A})) = r$ and $\dim(\text{Null}(\underline{A})) = n - r$ so $\text{Row}(\underline{A}) \cup \text{Null}(\underline{A}) = r + n - r = n$. That means any vector in $\mathbb{R}^n$ can be written as a linear combination of $\text{Row}(\underline{A})$ and $\text{Null}(\underline{A})$

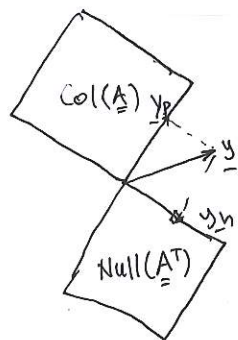


Example: $\underline{A} = \begin{bmatrix} 1 & 2 & 4 & 3 \\ 1 & 3 & 4 & 4 \\ 1 & 3 & 4 & 4 \\ 1 & 3 & 4 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 4 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} \underline{I} & \underline{N} \\ \underline{0} & \underline{0} \end{bmatrix}$ $\underline{N}^T = \begin{bmatrix} 4 & 1 \\ 0 & 1 \end{bmatrix}$

$\text{Row}(\underline{A}) = \left\{ \begin{bmatrix} 1 & 0 & 4 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} \right\} = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} \right\} \rightarrow \text{rows of } \begin{bmatrix} \underline{I} & \underline{N} \end{bmatrix}$
 $\underline{y} \in \text{Row}(\underline{A}) \Rightarrow \underline{y} = \begin{bmatrix} \underline{I} \\ \underline{N}^T \end{bmatrix} \underline{\alpha}$

$\text{Null}(\underline{A}) = \text{span} \left\{ \begin{bmatrix} -4 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\} \rightarrow \text{columns of } \begin{bmatrix} -\underline{N} \\ \underline{I} \end{bmatrix} = \underline{u} \in \text{Null}(\underline{A}) \Rightarrow \underline{u} = \begin{bmatrix} -\underline{N} \\ \underline{I} \end{bmatrix} \underline{\beta}$

$\underline{v}^T \underline{u} = \underline{\alpha}^T \begin{bmatrix} \underline{I} & \underline{N} \end{bmatrix} \begin{bmatrix} -\underline{N} \\ \underline{I} \end{bmatrix} \underline{\beta} = \underline{\alpha}^T (-\underline{N} + \underline{N}) \underline{\beta} = \underline{\alpha}^T \underline{0} \underline{\beta} = 0$



* Left nulls space, $\text{Null}(\underline{A}^T)$, is defined as the vectors $\underline{w}$ that satisfies the following equality. $\underline{w}^T \underline{A} = \underline{0} \Rightarrow \underline{A}^T \underline{w} = \underline{0}$

(i) We say $\underline{z} \in \text{Col}(\underline{A})$ if $\underline{z} = \underline{A}\underline{x}$ and $\underline{w} \in \text{Null}(\underline{A}^T)$ if $\underline{A}^T \underline{w} = \underline{0}$.

- Let $\underline{B} = \underline{A}^T$ and $\underline{E}$ is the reduced row echelon form of $\underline{B}$ that has the following format (we can always permute columns of $\underline{E}$ to obtain that format).

$\underline{E} = \begin{bmatrix} \underline{I}_r & \underline{M}_{r \times (m-r)} \\ \underline{0}_{(n-r) \times m} \end{bmatrix}_{n \times m} \Rightarrow \text{Col}(\underline{A}) = \text{span of columns of } \begin{bmatrix} \underline{I} \\ \underline{M}^T \end{bmatrix}, \underline{z} \in \text{Col}(\underline{A}) \Rightarrow \underline{z} = \begin{bmatrix} \underline{I} \\ \underline{M}^T \end{bmatrix} \underline{\theta}$
 $\text{Null}(\underline{A}^T) = \text{span of columns of } \begin{bmatrix} -\underline{M} \\ \underline{I} \end{bmatrix} \underline{w} \in \text{Null}(\underline{A}^T) = \underline{w} = \begin{bmatrix} -\underline{M} \\ \underline{I} \end{bmatrix} \underline{\theta}$

$\underline{z}^T \underline{w} = \underline{x}^T \begin{bmatrix} \underline{I} & \underline{M} \end{bmatrix} \begin{bmatrix} -\underline{M} \\ \underline{I} \end{bmatrix} \underline{\theta} = \underline{x}^T [-\underline{M} + \underline{M}] \underline{\theta} = \underline{x}^T [\underline{0}] \underline{\theta} = 0$

- Therefore any vector in $\text{Col}(\underline{A})$ is orthogonal to $\text{Null}(\underline{A}^T)$. Any vector $\underline{y} \in \mathbb{R}^m$ can be written as a linear combination of $\text{Col}(\underline{A})$ and $\text{Null}(\underline{A}^T)$

$\dim \{ \text{Null}(\underline{A}^T) \cup \text{Col}(\underline{A}) \} = (m - r) + r = m$

State Transition matrix (Examples from the example that is solved in last lecture)

Let the solution of the homogeneous system ($\underline{x}' = \underline{A}\underline{x}$) be $\underline{\phi}(t; t_0, \underline{x}_0)$ with initial condition $\underline{x}_0$.

Example: $\underline{A} = \begin{bmatrix} -4 & -2 \\ 3 & 1 \end{bmatrix}$ then the solution for initial condition $\underline{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is $\begin{bmatrix} 5e^{-2t} - 4e^{-t} \\ 6e^{-t} - 5e^{-2t} \end{bmatrix}$

then $\underline{\phi}(t; \underline{x}_0, \begin{bmatrix} 1 \\ 1 \end{bmatrix}) = \begin{bmatrix} 5e^{-2t} - 4e^{-t} \\ 6e^{-t} - 5e^{-2t} \end{bmatrix}$ is a two dimensional vector.

* If $\underline{x}_0 = \sum_{i=1}^n \alpha_i \underline{x}_i = \underline{X}_0 \underline{\alpha}$ then $\underline{x}(t) = \underline{\phi}(t; t_0, \underline{x}_0) = \sum \alpha_i \underline{\phi}(t; t_0, \underline{x}_i) = \begin{bmatrix} \underline{\phi}(t; t_0, \underline{x}_1) & \dots & \underline{\phi}(t; t_0, \underline{x}_n) \end{bmatrix} \underline{\alpha}$
 $\underline{\phi}(t; t_0, \underline{x}_0) = \underline{\phi}(t; t_0, \underline{X}_0) \underline{\alpha}$

$\underline{\phi}(t; t_0, \underline{x}_0)$ is the Fundamental Matrix with initial conditions $\underline{x}_0$.

$$\underline{\phi}(t; t_0, \underline{x}_0) = \underline{\phi}(t; t_0, \underline{x}_0) \underline{\alpha} = \underline{\phi}(t; t_0, \underline{x}_0) \underline{X}_0^{-1} \underline{x}_0 = \underline{\phi}(t; t_0) \underline{x}_0$$

$\underline{\phi}(t; t_0)$ is called state transition matrix so $\underline{\phi}(t; t_0, \underline{x}_0) = \underline{\phi}(t; t_0) \underline{x}_0$

$\underline{\phi}(t; t_0) = \underline{\phi}(t; t_0, \underline{I})$, Define $\underline{\phi}(t) = \underline{\phi}(t; 0) \rightarrow$ state transition matrix with $t_0 = 0$

Example: For $\underline{A} = \begin{bmatrix} -4 & -2 \\ 3 & 1 \end{bmatrix}$, $\underline{\phi}(t; t_0, \begin{bmatrix} 1 \\ 0 \end{bmatrix}) = \begin{bmatrix} 3e^{-2t} - 2e^{-t} \\ -3e^{-2t} + 3e^{-t} \end{bmatrix}$, $\underline{\phi}(t; t_0, \begin{bmatrix} 0 \\ 1 \end{bmatrix}) = \begin{bmatrix} 2e^{-2t} - 2e^{-t} \\ -2e^{-2t} + 3e^{-t} \end{bmatrix}$

$$\underline{\phi}(t; t_0, \underline{I}) = \underline{\phi}(t; t_0 = 0) = \underline{\phi}(t) = \begin{bmatrix} 3e^{-2t} - 2e^{-t} & 2e^{-2t} - 2e^{-t} \\ -3e^{-2t} + 3e^{-t} & -2e^{-2t} + 3e^{-t} \end{bmatrix}$$

- For the system $\underline{x}' = \underline{A}\underline{x}$ with constant coefficient matrix $\underline{A}$ we have

$$\mathcal{L}(\underline{x}' = \underline{A}\underline{x}) = s\underline{X}(s) - \underline{x}(0) = \underline{A}\underline{X}(s) \Rightarrow \underline{X}(s) = (s\underline{I} - \underline{A})^{-1} \underline{x}(0)$$

$$\text{then } \underline{\phi}(t) = \mathcal{L}^{-1}((s\underline{I} - \underline{A})^{-1}) = e^{\underline{A}t} \Rightarrow \underline{x}(t) = e^{\underline{A}t} \underline{x}(0) = \underline{\phi}(t; 0, \underline{x}(0))$$

- Compute state transition matrix for constant $\underline{A}$.

(i) Distinct eigenvalues or Linearly independent eigenvectors. $e^{\underline{A}t} = \underline{V} e^{\underline{\Lambda}t} \underline{V}^{-1}$ $\underline{V} = [\underline{v}_1 \dots \underline{v}_n]$ $\underline{v}_i$ eigenvector associated with λ_i

$$\text{let } \underline{x}(0) = \sum \alpha_i \underline{v}_i = \underline{V} \underline{\alpha} \Rightarrow \underline{\phi}(t; 0, \underline{x}(0)) = \underline{V} e^{\underline{\Lambda}t} \underline{V}^{-1} \underline{V} \underline{\alpha} = \sum \alpha_i e^{\lambda_i t} \underline{v}_i$$

(ii) Linearly dependent eigenvalues. $e^{\underline{A}t} = \underline{P} e^{\underline{J}t} \underline{P}^{-1}$ $\underline{P}$ generalized eigenvector matrix

for each Jordan block $e^{\underline{J}_{k \times k} t} = \begin{bmatrix} e^{\lambda_k t} & t e^{\lambda_k t} & \frac{t^2}{2} e^{\lambda_k t} & \dots & \frac{t^{n-1}}{(n-1)!} e^{\lambda_k t} \\ & e^{\lambda_k t} & t e^{\lambda_k t} & \dots & \frac{t^{n-2}}{(n-2)!} e^{\lambda_k t} \\ & & e^{\lambda_k t} & \dots & \frac{t^{n-3}}{(n-3)!} e^{\lambda_k t} \\ & & & \ddots & e^{\lambda_k t} \end{bmatrix}$ $\underline{J}$ Jordan form of $\underline{A}$

(iii) Use Cayley-Hamilton

(iv) Partial fraction expansion of $(s\underline{I} - \underline{A})^{-1}$ then take inverse Laplace transform.

$$* \underline{\phi}(t+t_0) = e^{\underline{A}(t+t_0)} = e^{\underline{A}t} \cdot e^{\underline{A}t_0} = \underline{\phi}(t) \underline{\phi}(t_0)$$

$$\text{③ } \underline{x}(t) = \underline{x}(t_0)$$

$$\underline{\phi}(t; 0, \underline{x}(0)) = e^{\underline{A}t} \underline{x}(0) \rightarrow \underline{\phi}(t+t_0; 0, \underline{x}(0)) = e^{\underline{A}(t+t_0)} \underline{x}(0) = e^{\underline{A}t} e^{\underline{A}t_0} \underline{x}(0) = e^{\underline{A}t} \underline{x}(t_0)$$

$$\underline{\phi}(t-t_0+t_0, 0, \underline{x}(0)) = e^{\underline{A}(t-t_0)} e^{\underline{A}t_0} \underline{x}(0) = e^{\underline{A}(t-t_0)} \underline{x}(t_0)$$

$$\underline{\phi}(t; t_0, \underline{x}(t_0)) = e^{\underline{A}(t-t_0)} \underline{x}(t_0)$$

$$\underline{\Phi}(t; 0, x(0)) = e^{A t} x(0) = e^{A(t-t_0)} e^{A t_0} x(0) = e^{A(t-t_0)} x(t_0) = \underline{\Phi}(t; t_0, x(t_0))$$

* If we have $x(0)$ for $t=0$ and $x(t_0)$ for $t=t_0$, then starting from $x(0)$ at $t=0$ has the same solution if start from $x(t_0)$ at t_0 .

* If at any time t_0 , the vector $x(t_0) = 0$ then $x(t) = 0$, so $x(t)$ can not be zero vector at any time if $x(t)$ is not identically zero at all times.

$$\underline{\Phi}(t; t_0) \underline{\Phi}(t_0; t_0) = \underline{\Phi}(t; t_0), \quad \boxed{\underline{\Phi}(t, t_0) = \underline{\Phi}^{-1}(t_0; t)} \quad \text{since } \underline{\Phi}(t, t_0) \underline{\Phi}(t_0, t) = \underline{\Phi}(t; t) = \underline{I}$$

$$\underline{\Phi}(t-t_0; 0) = \underline{\Phi}(t; t_0) \quad \underline{\Phi}(t; 0, x(0)) = \underline{\Phi}(t; t_0, x(t_0))$$

$$\underline{\Phi}(t; 0) x(0) = \underline{\Phi}(t; t_0) x(t_0) = \underline{\Phi}(t; t_0) \underline{\Phi}(t_0; 0) x(0)$$

$$\underline{\Phi}(t; 0) \underline{\Phi}(0; t_0) = \underline{\Phi}(t-t_0; 0) = \underline{\Phi}(t; t_0)$$

Discrete Convolution:

$$(1) y[k] = \sum_{m=0}^k x[m] h[k-m] \Leftrightarrow Y(z) = X(z) H(z) \quad x[k] = h[k] = 0 \text{ for } k < 0$$

$$(2) y(t) = \int_0^t x(\tau) h(t-\tau) d\tau \Leftrightarrow X(s) H(s) = Y(s) \quad x(t) = h(t) = 0 \text{ for } t < 0$$

$$\begin{aligned} \text{Proof: } Y(z) &= \sum_k Y[k] z^{-k} = \sum_{k=0}^{\infty} \sum_{m=0}^k x[m] h[k-m] z^{-k} = \sum_{k=0}^{\infty} \sum_{m=0}^{\infty} x[m] h[k-m] z^{-k} \\ &\xrightarrow{\text{since } h[k]=0 \text{ for } k < 0} \sum_{m=0}^{\infty} x[m] \sum_{k=0}^{\infty} h[k-m] z^{-k} = \sum_{m=0}^{\infty} x[m] \sum_{n=-m}^{\infty} h[n] z^{-(m+n)} = \sum_{m=0}^{\infty} x[m] z^{-m} \sum_{n=0}^{\infty} h[n] z^{-n} \\ &\xrightarrow{\text{since } \sum_{n=-m}^{-1} h[n] z^{-(m+n)} = 0 \text{ for } k < 0} \left(\sum_{m=0}^{\infty} x[m] z^{-m} \right) \left(\sum_{n=0}^{\infty} h[n] z^{-n} \right) = X(z) H(z) \end{aligned}$$

$$(2) Y(s) = \int_0^{\infty} \int_0^t x(\tau) h(t-\tau) d\tau dt = \int_0^{\infty} \int_0^{\infty} x(\tau) h(t-\tau) d\tau dt = \int_0^{\infty} x(\tau) \int_0^{\infty} h(t-\tau) e^{-st} dt d\tau$$

$$= \int_0^{\infty} x(\tau) \int_0^{\infty} h(t-\tau) e^{-st} dt d\tau = \int_0^{\infty} x(\tau) \int_{-\tau}^{\infty} h(u) e^{-s(\tau+u)} du d\tau$$

$$= \int_0^{\infty} x(\tau) e^{-s\tau} d\tau \cdot \int_0^{\infty} h(u) e^{-su} du$$

$$\int_{-\tau}^0 h(u) e^{-s(\tau+u)} du = 0 \quad \text{since } h(t) = 0 \text{ for } t < 0.$$

State Transition

Example: Find zero-input and zero-state solutions for the following system.

$$\underline{x}' = \begin{bmatrix} -4 & -2 \\ 3 & 1 \end{bmatrix} \underline{x} + \begin{bmatrix} e^{-2t} \\ e^{-t} \end{bmatrix} = \underline{A} \underline{x} + \underline{f}$$

(i) Find the Laplace transform of $\underline{x}(s)$.

$$\mathcal{L}(\underline{x}') = s\underline{X}(s) - \underline{x}(0) \quad \mathcal{L}(\underline{x}) = \underline{X}(s) \quad \mathcal{L}(\underline{f}) = \underline{F}(s)$$

$$s\underline{X}(s) - \underline{x}(0) = \underline{A} \underline{X}(s) + \underline{F}(s)$$

$$(s\underline{I} - \underline{A})\underline{X}(s) = \underline{x}(0) + \underline{F}(s) \Rightarrow \underline{X}(s) = \underbrace{(s\underline{I} - \underline{A})^{-1}}_{\underline{ZI}} \underline{x}(0) + \underbrace{(s\underline{I} - \underline{A})^{-1}}_{\underline{ZS}} \underline{F}(s) \rightarrow \underline{F}(s) = \begin{bmatrix} 1/s+2 \\ 1/s+1 \end{bmatrix}$$

(iii) Find the inverse Laplace transform of $\underline{X}_{ZI}(s)$ and $\underline{X}_{ZS}(s)$.

$$\underline{X}_{ZI}(s) = (s\underline{I} - \underline{A})^{-1} \underline{x}_0 \xrightarrow{\mathcal{L}^{-1}} \underline{x}_{ZI}(t) = e^{\underline{A}t} \underline{x}_0 = \begin{bmatrix} 1 & 2 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} e^{-2t} \\ e^{-t} \end{bmatrix} \begin{bmatrix} 3 & 2 \\ -1 & -1 \end{bmatrix}$$

State transition matrix $\lambda_1, \lambda_2 = (-2, -1)$

$$e^{\underline{A}t} = \underline{V} e^{\underline{D}t} \underline{V}^{-1} \quad \underline{V} = \begin{bmatrix} 1 & 2 \\ -1 & -3 \end{bmatrix} \rightarrow \underline{V}^{-1} = \begin{bmatrix} 3 & 2 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -1 & -3 \end{bmatrix} \begin{bmatrix} 3e^{-2t} & 2e^{-2t} \\ -e^{-t} & -e^{-t} \end{bmatrix}$$

$$= \begin{bmatrix} 3e^{-2t} - 2e^{-t} & 2e^{-2t} - 2e^{-t} \\ -3e^{-2t} + 3e^{-t} & -2e^{-2t} + 3e^{-t} \end{bmatrix}$$

if $\underline{x}_0 = \underline{V} \underline{\alpha}$ $e^{\underline{A}t} \underline{x}_0 = \underline{V} e^{\underline{D}t} \underline{V}^{-1} \underline{x}_0 = \underline{V} e^{\underline{D}t} \underline{V}^{-1} \underline{V} \underline{\alpha} = \underline{V} e^{\underline{D}t} \underline{\alpha} = \sum \alpha_i e^{\lambda_i t} \underline{v}_i$

(b) Partial fraction expansion

$$(s\underline{I} - \underline{A})^{-1} = \begin{bmatrix} s+4 & 2 \\ -3 & s-1 \end{bmatrix}^{-1} = \frac{1}{(s+2)(s+1)} \begin{bmatrix} s-1 & -2 \\ 3 & s+4 \end{bmatrix} = \frac{1}{s+1} \begin{bmatrix} -2 & -2 \\ 3 & 3 \end{bmatrix} + \frac{1}{s+2} \begin{bmatrix} 3 & 2 \\ -3 & -2 \end{bmatrix} \quad (\text{same})$$

$$= e^{-t} \begin{bmatrix} -2 & -2 \\ 3 & 3 \end{bmatrix} + e^{-2t} \begin{bmatrix} 3 & 2 \\ -3 & -2 \end{bmatrix}$$

2S (a) Partial fraction expansion.

$$\underline{X}_{ZS}(s) = (s\underline{I} - \underline{A})^{-1} \underline{F}(s) = \frac{1}{(s+2)(s+1)} \begin{bmatrix} s-1 & -2 \\ 3 & s+4 \end{bmatrix} \begin{bmatrix} 1/s+2 \\ 1/s+1 \end{bmatrix} = \frac{1}{(s+1)^2(s+2)^2} \begin{bmatrix} s^2 - 2s - 5 \\ s^2 + 9s + 11 \end{bmatrix}$$

$$= \frac{1}{s+1} \begin{bmatrix} A \\ B \end{bmatrix} + \frac{1}{(s+1)^2} \begin{bmatrix} F \\ G \end{bmatrix} + \frac{1}{s+2} \begin{bmatrix} C \\ D \end{bmatrix} + \frac{1}{(s+2)^2} \begin{bmatrix} H \\ J \end{bmatrix} =$$

$$\begin{bmatrix} H \\ J \end{bmatrix} = (s+2)^2 \underline{X}_{ZS}(s) \Big|_{s=-2} = \frac{1}{(s+1)^2} \begin{bmatrix} s^2 - 2s - 5 \\ s^2 + 9s + 11 \end{bmatrix} \Big|_{s=-2} = \begin{bmatrix} 3 \\ -3 \end{bmatrix}, \quad \begin{bmatrix} F \\ G \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} C \\ D \end{bmatrix} = \frac{d}{ds} (s+2)^2 \underline{X}_{ZS}(s) \Big|_{s=-2} = \frac{d}{ds} \frac{1}{(s+1)^2} \begin{bmatrix} s^2 - 2s - 5 \\ s^2 + 9s + 11 \end{bmatrix} \Big|_{s=-2} = \frac{\begin{bmatrix} 2s-2 \\ 2s+9 \end{bmatrix} (s+1)^2 + 2(s+1) \begin{bmatrix} s^2 - 2s - 5 \\ s^2 + 9s + 11 \end{bmatrix}}{(s+1)^4} \Big|_{s=-2} = \frac{\begin{bmatrix} -6 \\ 5 \end{bmatrix} + \begin{bmatrix} +6 \\ -6 \end{bmatrix}}{1} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} A \\ B \end{bmatrix} = \frac{d}{ds} (s+1)^2 \underline{X}_{ZS}(s) \Big|_{s=-1} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$x_{zs}(t) = e^{-t} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + t e^{-t} \begin{bmatrix} -2 \\ 3 \end{bmatrix} + e^{-2t} \begin{bmatrix} 0 \\ -1 \end{bmatrix} + t e^{-2t} \begin{bmatrix} 3 \\ -3 \end{bmatrix}$$

(b) Convolution:

$$x_{zs}(t) = \int_0^t \underline{\Phi}(t;\tau) \underline{f}(\tau) d\tau = \int_0^t \left(e^{-(t-\tau)} \underline{X}_1 + e^{-2(t-\tau)} \underline{X}_2 \right) \begin{bmatrix} e^{-2\tau} \\ e^{-\tau} \end{bmatrix} d\tau$$

$$= \int_0^t \underline{X}_1 \begin{bmatrix} e^{-(t-\tau+2\tau)} \\ e^{-(t-\tau+\tau)} \end{bmatrix} d\tau + \int_0^t \underline{X}_2 \begin{bmatrix} e^{-2(t-\tau+\tau)} \\ e^{-(t-2\tau+\tau)} \end{bmatrix} d\tau$$

$$= e^{-t} \underline{X}_1 \int_0^t \begin{bmatrix} e^{-\tau} \\ 1 \end{bmatrix} d\tau + e^{-2t} \underline{X}_2 \int_0^t \begin{bmatrix} 1 \\ e^{\tau} \end{bmatrix} d\tau$$

$$= e^{-t} \underline{X}_1 \begin{bmatrix} -e^{-\tau} \Big|_0^t \\ t \end{bmatrix} + e^{-2t} \underline{X}_2 \begin{bmatrix} t \\ e^{\tau} \Big|_0^t \end{bmatrix} d\tau$$

$$= e^{-t} \underline{X}_1 \begin{bmatrix} 1-e^{-t} \\ t \end{bmatrix} + e^{-2t} \underline{X}_2 \begin{bmatrix} t \\ e^t - 1 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & -2 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} e^{-t} - e^{-2t} \\ t e^{-t} \end{bmatrix} + \begin{bmatrix} 3 & 2 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} t e^{-2t} \\ e^{-t} - e^{-2t} \end{bmatrix}$$

$$= \begin{bmatrix} 2e^{-2t} - 2e^{-t} - 2te^{-t} + 3te^{-2t} + 2e^{-t} - 2e^{-2t} \\ 3e^{-t} - 3e^{-2t} + 3te^{-t} - 3te^{-2t} + 2e^{-2t} - 2e^{-t} \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 1 \end{bmatrix} e^{-t} + t e^{-t} \begin{bmatrix} -2 \\ 3 \end{bmatrix} + e^{-2t} \begin{bmatrix} 0 \\ -1 \end{bmatrix} + t e^{-t} \begin{bmatrix} 3 \\ -3 \end{bmatrix} \rightarrow \text{same!}$$